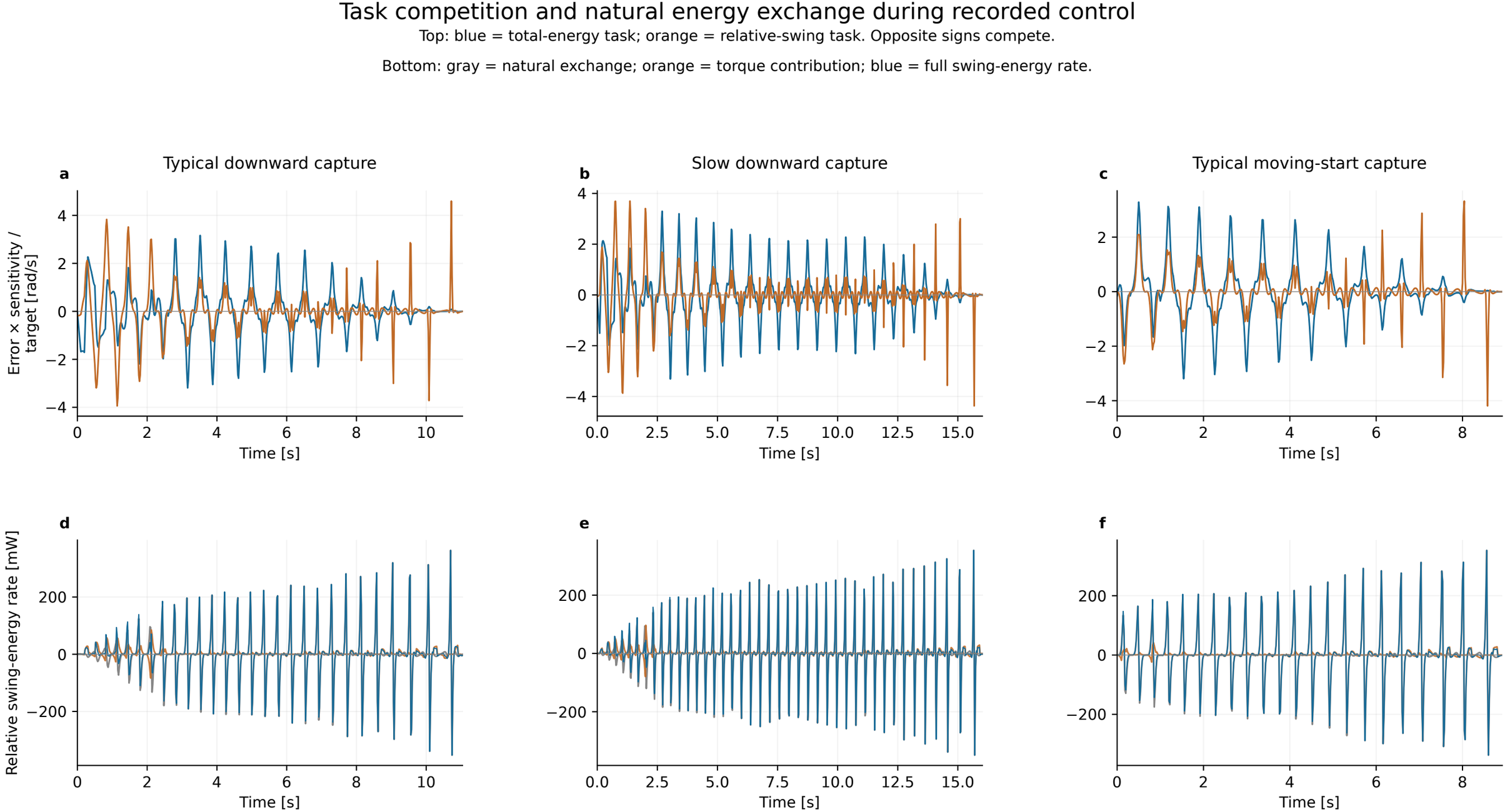

Task competition and natural energy exchange during recorded control
Top: blue = total-energy task; orange = relative-swing task. Opposite signs compete.
Bottom: gray = natural exchange; orange = torque contribution; blue = full swing-energy rate.
a
Typical downward capture
b
Slow downward capture
c
Typical moving-start capture
Error × sensitivity / target [rad/s]
d
e
f
Relative swing-energy rate [mW]
Time [s]